How UI and UX Change in the Agentic AI Era

A deep research report comparing pre-AI human-operated software with agentic AI systems, with implications for approval interfaces

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Executive summary

The central UX shift in the agentic era is a move from interfaces built for direct human operation to interfaces built for delegation, supervision, and bounded autonomy. In the pre-AI era, the main question was: how do we help a person complete a workflow? In the agentic era, the new question is: how do we help a person safely allocate decision rights, inspect what the system intends to do, constrain it, monitor it, and take back control when needed?

- The interaction unit shifts from screens, rows, and commands to goals, plans, constraints, and execution state.
- Approval shifts from final-output review to plan review, impact preview, policy checks, and evidence inspection.
- Control shifts from submit/cancel/undo to approve with limits, request re-plan, pause, stop, rollback, and manual takeover.
- Governance becomes an interface concern - risk tiers, agent identity, provenance, and auditability must be visible at decision time.
- Traditional forms and email approvals still work for low-risk, short, deterministic, and reversible actions; richer supervisory consoles are needed as autonomy, duration, and consequence increase.

Research base

This report synthesizes foundational HCI work on direct manipulation, mixed-initiative interfaces, and supervisory control with recent AI-specific design guidance, governance frameworks, and enterprise human-in-the-loop implementation patterns from 2024-2026.

1. Historical frame: from direct manipulation to delegation

For most of the software era, mainstream UI/UX assumed a human operator using deterministic tools. The design ideal was direct manipulation: visible objects, understandable actions, and a strong sense of control and predictability. Users learned the system, drove each step, and generally remained the obvious source of agency in the workflow. [1][2][3]

The possibility of more delegated interaction is not new. In 1997, the debate between Ben Shneiderman and Pattie Maes framed the issue as direct manipulation versus interface agents. Even then, the core tension was clear: humans value control, predictability, and responsibility, yet growing complexity makes some delegation attractive. Horvitz's mixed-initiative research and later work on delegation interfaces proposed a middle path, where humans and intelligent services collaborate and the human remains able to invoke, shape, and supervise automation rather than being replaced by it. [1][2][3]

Agentic AI makes that middle path a day-to-day product problem. The system is no longer only helping a user fill a form, retrieve information, or automate a fixed macro. It may interpret goals, call tools, act across systems, re-plan in flight, and create external side-effects. As a result, the interface is no longer just where work gets done. It is where decision rights are delegated, bounded, observed, and reclaimed. [10][11][16]

> "*Agency isn't a feature - it's a transfer of decision rights.*"
>
> Industry framing from McKinsey, 2026 [16]

2. Why interfaces change now

Three developments make the shift urgent. First, current agent systems can work on long-running and complex tasks, and those tasks can produce irreversible real-world side-effects. Microsoft Research's Magentic-UI report treats this as a fundamental distinction between agents and traditional AI models. Second, formal governance expectations are becoming explicit. NIST identifies Human-AI Configuration as a risk category that includes automation bias, over-reliance, and anthropomorphization. The EU AI Act requires human oversight, including human-machine interface tools that allow people to monitor, interpret, override, or safely stop high-risk systems. Third, adoption is racing ahead of mature oversight. The World Economic Forum notes that organizations are moving from prototypes toward deployment even as governance practices remain uneven. [8][9][10][11][15]

Longer task horizons	Higher consequence	Stronger oversight pressure
Agents can handle multi-step work that lasts minutes or hours, not only instant suggestions.	Actions may touch money, data, infrastructure, customer communications, or identity systems.	Risk frameworks and regulations now expect monitoring, checkpoints, overrides, and reconstructable traces.

Comparison matrix: pre-AI software vs. agentic systems

The table below compresses the biggest shift. Traditional enterprise UX treated software as an obedient instrument. Agentic UX treats software as an actor that can plan and execute, so the interface must support delegation, supervision, and intervention rather than only direct operation. [1][2][3][4][8][10][11]

Dimension	Pre-AI / human-operated era	Agentic AI era
Mental model	A tool the human directly operates.	A delegated actor or service that proposes, decides, and can act within limits.
Primary user job	Complete fields, click controls, and drive each step.	Set intent, scope, and guardrails; inspect plans; supervise execution.
Unit of interaction	Screen, form, row, command, transaction.	Goal, plan, subtask, policy checkpoint, execution state.
System behavior	Mostly deterministic and workflow-bound.	Probabilistic, adaptive, and able to re-plan under changing context.
What gets reviewed	Final record or final action.	Proposed plan, evidence, expected impact, policy results, and side-effects.
State visibility	Current page state and database state.	Long-running task state, tool calls, dependencies, partial completions, and re-approvals.
Control surface	Submit, cancel, edit, undo.	Approve with limits, request re-plan, pause, stop, rollback, take over manually.
Failure model	Validation errors, crashes, and permissions issues.	Uncertainty, confabulation, over-reliance, hidden autonomy, and irreversible side-effects.
Governance	Permissions and after-the-fact audit logs.	Risk-tiered autonomy, human oversight, identity binding, live monitoring, and reconstructable traces.
Success metrics	Speed, completion rate, and data accuracy.	Safe autonomy rate, trust calibration, override quality, policy compliance, and audit completeness.

3. Seven design shifts that redefine UI/UX

3.1 From commands to goals plus constraints

Generative-AI research increasingly describes interaction as intent-based rather than command-based. Agentic systems extend this from content generation to delegated execution. That means the interface must capture not only what the user wants, but also the bounds within which the system may pursue it: budget, scope, tools, time window, retry count, escalation path, and conditions that require re-approval. [7][10]

3.2 From output review to plan, evidence, and impact review

In a classic workflow, the human often reviews the final record or final action. In an agentic workflow, reviewing only the end result is too late. The review object becomes the proposed plan, the evidence behind it, the expected side-effects, and the policies it will trigger. This is why co-planning, action approval, and answer verification appear as explicit interaction mechanisms in Magentic-UI, and why enterprise human-approval patterns increasingly pause execution at checkpoints. [11][12][13][14]

3.3 From one-shot interactions to supervisory control loops

Pre-AI UX often optimized for a single loop: input, validate, submit, confirm. Supervisory control requires a broader loop: propose, inspect, approve, execute, monitor, intervene, and learn. Earlier human-AI guidance already emphasized timing services by context, supporting efficient invocation, and allowing correction. Agentic systems make those ideas structural rather than optional. [2][3][4]

3.4 From hidden policy to visible guardrails

In conventional enterprise products, policy often lives behind the interface as business logic. In agentic systems, policy must become legible. Reviewers need to know which safeguards passed, which failed, which were only warnings, and who can override them. The EU AI Act and Singapore's agentic governance framework both push toward oversight that is proportional to risk, autonomy, and context of use. [9][10]

3.5 From static permissions to bounded autonomy

A permission like 'can approve invoices' is not precise enough for an agent. Agentic approval means authorizing a bounded operating envelope: amount ceiling, allowed systems, data classes, retry policy, and whether a plan change invalidates prior approval. The decision is less 'yes or no' than 'yes, within these constraints.' [10][16]

3.6 From system messages to live telemetry and interruptibility

Traditional status messages report that something finished or failed. Agentic UX needs richer telemetry: step progress, tool usage, checkpoints, deviations, re-plans, and the ability to interrupt safely. Article 14 of the EU AI Act explicitly highlights the ability to override, reverse, or interrupt a system through a stop button or similar procedure that brings it to a safe state. [9][11]

3.7 From usability-only metrics to trust, controllability, and reconstructability

Completion time and task success still matter, but they are no longer enough. The performance of a human-agent system also depends on whether trust is calibrated rather than inflated, whether overrides happen early enough to prevent harm, whether policies catch the right edge cases, and whether the organization can reconstruct each decision and action end-to-end. NIST, the World Economic Forum, and current agent-governance playbooks all push toward this broader measurement frame. [8][15][16]

What changes in approval UX

Conventional approval interfaces usually assume the record already exists and the human is deciding whether to let a fixed action proceed. Agentic approval interfaces must supervise a system that may still be planning, may call tools, may re-plan after new evidence, and may generate side-effects while executing. [9][10][11][12][13][14]

Traditional pattern	Agentic replacement	Why it changes
Row-level selection toolbar	Risk-prioritized queue with task cards and autonomy level	Approvers must triage tasks by consequence, ambiguity, and time sensitivity - not only by count.
Confirmation modal	Dry-run impact summary and policy/evidence preview	Humans need to inspect likely side-effects before the action happens.
Approve / reject	Approve plan, approve with limits, request re-plan, escalate	Binary decisions are too coarse when the agent can be constrained or redirected.
Status badge	Live execution timeline with checkpoints and re-approval triggers	Task state is no longer a static record status; it is a process with partial completion and change over time.
Permissions page	Inline constraint editor for tools, systems, amount, time window, retries	The decision is not only whether the agent may act, but under what boundaries it may act.
Audit log	Replayable action trail with agent identity, owner, rationale, and outcomes	Supervision quality depends on reconstructing what happened and who was accountable.

Conceptual wireframe: from bulk approval to agent supervision

The figure below is an author-generated mockup that synthesizes the research into a concrete interface pattern. It is not a product screenshot. The key change is visible immediately: the center of gravity is no longer a table of rows plus a bulk action bar. It is a supervision workspace for plan review, bounded approval, execution monitoring, and auditability. [9][10][11][12][13][14]

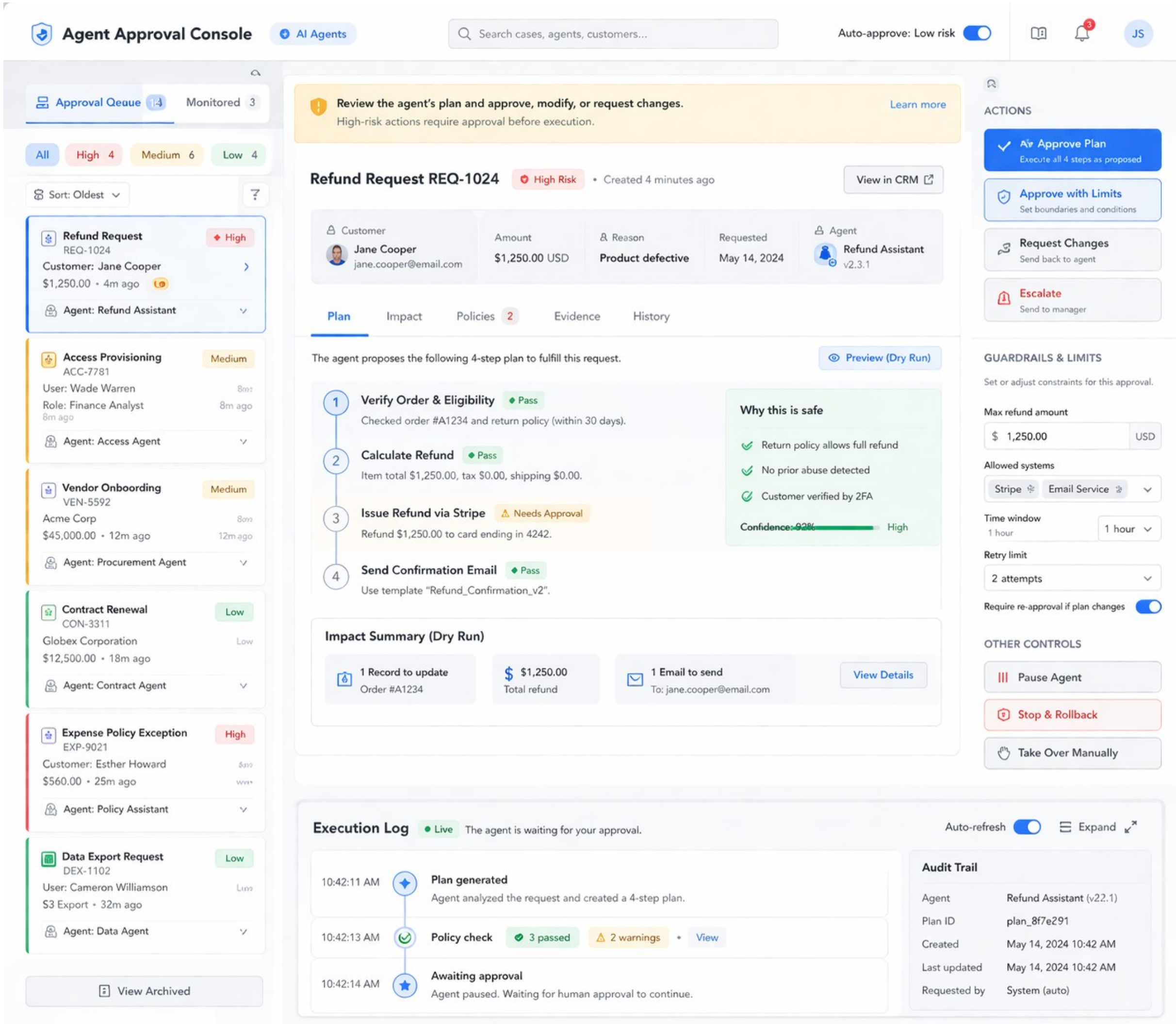

Figure 1. Conceptual wireframe of an agentic AI approval console (author-generated mockup).

Visible region	Why it exists in agentic systems
Queue and risk filters	Supervisors need triage based on consequence, urgency, and autonomy level rather than a flat inbox.
Plan view	The human is approving a proposed sequence of actions, not only an end-state row value.
Impact summary	A dry run externalizes side-effects before commitment: records changed, money moved, messages sent, or systems touched.
Policies and evidence	Guardrails, rationale, and uncertainty must be visible at decision time so approval is informed rather than ceremonial.
Constraints and action rail	Approve with limits, request changes, pause, rollback, or take over manually - these are supervisory controls, not ordinary form actions.
Execution log and audit trail	Because tasks are long-running and may re-plan, the system must make behavior reconstructable end-to-end.

4. Design principles for product teams

- Default to bounded autonomy, not hidden autonomy. Make the agent's scope explicit before it acts. [9][10]
- Separate plan approval from execution approval when side-effects are material. One click should not silently authorize every downstream action. [11][12][13]
- Make policy legible at decision time. Show guardrails, pass/fail results, warnings, and who can override them. [9][10]
- Design for interruption and recovery. Pause, stop, rollback, and manual takeover are first-class controls for high-stakes tasks. [9][11]
- Use progressive disclosure for rationale. Provide evidence summaries and relevant context instead of overwhelming users with raw traces or faux certainty. [4][5][6]
- Calibrate trust. Show limitations, uncertainty, history, and missing information so users neither over-trust nor under-use the system. [4][5][8]
- Bind every action to identity and ownership. Reviewers should know which agent acted, under whose authority, with what tool access. [10][16]
- Measure supervision quality. Track not only throughput, but also override quality, near misses, policy hit rates, and audit completeness. [8][15][16]

Where the old UI still works

Form-based approvals and even email approvals remain appropriate when the action is short-lived, deterministic, low-risk, and fully reversible. Oracle's current human-in-the-loop tooling, for example, still treats forms as the interface for human interaction. The need for a full supervisory console grows with autonomy, ambiguity, duration, and consequence.

A practical maturity model

Not every team needs a full supervisory console on day one. A better path is to raise autonomy only as quickly as monitoring, controls, and accountability improve. [10][11][12][13][14][15][16]

Level	UX shape	Required controls
M0 - Human-operated workflow	Deterministic forms, dashboards, and workflow queues.	Standard permissions, validation, and undo.
M1 - AI assistant	AI suggests or drafts; human still executes the action.	Clear capabilities/limits, editable suggestions, source checking.
M2 - Bounded agent with checkpoints	Agent proposes actions and pauses for approval at specific gates.	Structured action payloads, dry runs, audit trail, timeout behavior.
M3 - Supervisory console	Human supervises live execution and can pause, stop, or re-scope the task.	Plan view, evidence/policy panels, constraints, takeover, rollback.
M4 - Governed multi-agent operations	Multiple agents coordinate within risk-tiered autonomy across systems.	Cross-agent observability, identity binding, policy orchestration, replay, containment.

5. Conclusion

The deepest UI/UX change in the agentic era is not that interfaces become more conversational. It is that interfaces become operational control surfaces for delegated software actors. Pre-AI UX primarily optimized direct manipulation of predictable tools. Agentic UX must optimize bounded delegation, meaningful oversight, safe interruption, and end-to-end accountability. [1][2][3][9][10][11]

In practice, that means approval UIs evolve from row-level batch actions into supervision consoles: they show plans rather than only outcomes, constraints rather than only permissions, live execution rather than only final status, and audit trails rather than only event crumbs. The design challenge is no longer just usability. It is governance made tangible at the moment of interaction. [8][10][11][15][16]

For product teams, the right question is not 'How do we add an AI button to the existing workflow?' It is 'What decision rights are being transferred, what guardrails must be visible in the interface, and how does the human stay effectively accountable as the system becomes more autonomous?' [4][5][10][16]

References

[1] Shneiderman, B., & Maes, P. Direct Manipulation vs Interface Agents. Interactions debate, 1997.
https://www.lri.fr/~mbl/ENS/FONDIHM/2013/papers/ShneidermanMaes-Interactions97.pdf

[2] Horvitz, E. Principles of Mixed-Initiative User Interfaces, 1999.
https://erichorvitz.com/chi99horvitz.pdf

[3] Miller, C. A., & Parasuraman, R. Designing for Flexible Interaction Between Humans and Automation: Delegation Interfaces for Supervisory Control, 2007.
https://www.sift.net/sites/default/files/publications/MP-HumanFactors-2007.pdf

[4] Amershi, S., et al. Guidelines for Human-AI Interaction. CHI 2019.
https://www.microsoft.com/en-us/research/wp-content/uploads/2019/01/Guidelines-for-Human-AI-Interaction-camera-ready.pdf

[5] Google PAIR. People + AI Guidebook, third edition updated April 2025.
https://pair.withgoogle.com/guidebook/

[6] Weisz, J. D., et al. Design Principles for Generative AI Applications. CHI 2024.
https://doi.org/10.1145/3613904.3642466

[7] Lin, L., et al. Design Guidelines for Human-Generative AI Interaction. HCII 2025.
https://link.springer.com/chapter/10.1007/978-3-031-93718-7_15

[8] NIST. Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (AI 600-1), 2024.
https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

[9] EU AI Act Service Desk. Article 14: Human Oversight.
https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-14

[10] Infocomm Media Development Authority (Singapore). Model AI Governance Framework for Agentic AI, 2025.
https://www.imda.gov.sg/-/media/imda/files/about/emerging-tech-and-research/artificial-intelligence/mgf-for-agentic-ai.pdf

[11] Microsoft Research. Magentic-UI: Towards Human-in-the-loop Agentic Systems, 2025.
https://www.microsoft.com/en-us/research/wp-content/uploads/2025/07/magentic-ui-report.pdf

[12] Microsoft Community Hub. Multi-agent Workflow with Human Approval using Agent Framework, 2025.
https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/multi-agent-workflow-with-human-approval-using-agent-framework/4465927

[13] GitHub / Microsoft. Human Oversight for AI Agents solution accelerator, 2026.
https://github.com/microsoft/agents-humanoversight

[14] Oracle Integration. Human in the Loop for Agentic AI in Oracle Integration, 2026.
https://docs.oracle.com/en/cloud/paas/application-integration/human-loop/human-loop-agentic-ai-oracle-integration.html

[15] World Economic Forum. AI Agents in Action: Foundations for Evaluation and Governance, 2025.
https://www.weforum.org/publications/ai-agents-in-action-foundations-for-evaluation-and-governance/

[16] McKinsey. Trust in the age of agents, 2026.
https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/trust-in-the-age-of-agents